THE STACK OF FINITE TYPE SPACES IS A STACK IN GROUPOIDS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file quot.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- Slogan bootstrap-section-applications (canonical tag 04SJ). cited at proof lines 3575-3576. Any fppf descent datum for algebraic spaces is effective.
- Lemma examples-stacks-lemma-stack-of-finite-type-spaces (canonical tag 04UD). cited at proof line 3579. The finite type case of condition (3).
- Section examples-stacks-section-stack-in-groupoids-of-finite-type-spaces (canonical tag 04UH). cited at proof line 3581. Where the finite type case was established.
- Definition stacks-definition-stack-in-groupoids (canonical tag 02ZI). cited at proof line 3584. Conditions (1), (2), (3) being verified.
- Lemma lemma-spaces-fibred-in-groupoids (canonical tag 0D1E). cited at proof line 3587. Condition (1): fibred in groupoids.
- Lemma lemma-spaces-diagonal (canonical tag 0D1F). cited at proof line 3591. Condition (2) in the fp,flat,proper case.
- Lemma examples-stacks-lemma-pre-stack-of-spaces (canonical tag 04UA). cited at proof lines 3594-3595. Condition (2) in the finite type case.
- Lemma bootstrap-lemma-descend-algebraic-space (canonical tag 0ADV). cited at proof line 3612. Condition (3): the descended algebraic space exists (part (2)).
- Lemma spaces-descent-lemma-descending-property-finite-type (canonical tag 041U). cited at proof line 3614. X -> U is of finite type.
- Lemma spaces-descent-lemma-descending-property-finite-presentation (canonical tag 041V). cited at proof line 3619. fp,flat,proper case: finite presentation descends.
- Lemma spaces-descent-lemma-descending-property-flat (canonical tag 041W). cited at proof line 3620. fp,flat,proper case: flatness descends.
- Lemma spaces-descent-lemma-descending-property-proper (canonical tag 0422). cited at proof line 3621. fp,flat,proper case: properness descends.

Source reference. Original section 04SJ: sources/stacks-chow/tags.txt, source lines 6222–6222. Cited at proof lines 3575-3576: the slogan underlying the whole lemma; load-bearing.

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Source reference. Original lemma 04UD: sources/stacks-chow/tags.txt, source lines 6286–6286. Cited at proof line 3579: the primary reference for condition (3) in the finite type case; load-bearing.

Source reference. Original section 04UH: sources/stacks-chow/tags.txt, source lines 6290–6290. Cited at proof line 3581: where the lemma was previously seen; load-bearing.

Source reference. Original definition 02ZI: sources/stacks-chow/tags.txt, source lines 3913–3913. Cited at proof line 3584: the three conditions being checked; load-bearing.

Source reference. Original lemma 0D1E: sources/stacks-chow/tags.txt, source lines 16344–16344. Cited at proof line 3587: condition (1); load-bearing.

Source reference. Original lemma 0D1F: sources/stacks-chow/tags.txt, source lines 16345–16345. Cited at proof line 3591: condition (2) in the fp,flat,proper case; load-bearing.

Source reference. Original lemma 04UA: sources/stacks-chow/tags.txt, source lines 6283–6283. Cited at proof lines 3594-3595: condition (2) in the finite type case; load-bearing.

Source reference. Original lemma 0ADV: sources/stacks-chow/tags.txt, source lines 13103–13103. Cited at proof line 3612: condition (3), the descended space X exists; load-bearing.

Source reference. Original lemma 041U: sources/stacks-chow/tags.txt, source lines 5305–5305. Cited at proof line 3614: X -> U is of finite type; load-bearing.

Source reference. Original lemma 041V: sources/stacks-chow/tags.txt, source lines 5306–5306. Cited at proof line 3619: fp,flat,proper case; load-bearing.

Source reference. Original lemma 041W: sources/stacks-chow/tags.txt, source lines 5307–5307. Cited at proof line 3620: fp,flat,proper case; load-bearing.

Source reference. Original lemma 0422: sources/stacks-chow/tags.txt, source lines 5313–5313. Cited at proof line 3621: fp,flat,proper case; load-bearing.

Lemma 0.2. *The category $\mathcal{S}paces'_{ft}$ is a stack in groupoids over Sch_{fppf}. The same is true for $\mathcal{S}paces'_{fp,flat,proper}$.*

Proof. The reason this lemma holds is the slogan: any fppf descent datum for algebraic spaces is effective, see Bootstrap, Section 04SJ. More precisely, the lemma for $\mathcal{S}paces'_{ft}$ follows from Examples of Stacks, Lemma 04UD as we saw in Examples of Stacks, Section 04UH. However, let us review the proof. We need to check conditions (1), (2), and (3) of Stacks, Definition 02ZI.

Property (1) we have seen in Lemma 0D1E.

Property (2) follows from Lemma 0D1F in the case of $\mathcal{S}paces'_{fp,flat,proper}$. In the case of $\mathcal{S}paces'_{ft}$ it follows from Examples of Stacks, Lemma 04UA (and this is really the "correct" reference).

Condition (3) for $\mathcal{S}paces'_{ft}$ is checked as follows. Suppose given

(1) an fppf covering $\{U_i \to U\}_{i\in I}$ in Sch_{fppf},
(2) for each $i \in I$ an algebraic space X_i of finite type over U_i, and

(3) for each $i, j \in I$ an isomorphism $\varphi_{ij} : X_i \times_U U_j \to U_i \times_U X_j$ of algebraic spaces over $U_i \times_U U_j$ satisfying the cocycle condition over $U_i \times_U U_j \times_U U_k$.

We have to show there exists an algebraic space X of finite type over U and isomorphisms $X_{U_i} \cong X_i$ over U_i recovering the isomorphisms φ_{ij}. This follows from Bootstrap, Lemma 0ADV part (2). By Descent on Spaces, Lemma 041U we see that $X \to U$ is of finite type. In the case of $\mathcal{S}paces'_{fp,flat,proper}$ one additionally uses Descent on Spaces, Lemma 041V, 041W, and 0422 in the last step. □